SIMATS SCHOOL OF ENGINEERING

Approved By AICTE | IET-UK Accreditation

Course Code: DSA0311	Course Name: Natural Language Processing
Branch: AI&DS	Year : 2024-2028
Name	A.Abhishek Reddy
Rg no	192424192

Assignment 3

1. Draw tree structures for the following ATIS sentences: (10M) (CO3) (BLT4)

a. Does American airlines have a flight between five a.m. and six a.m.

b. I would like to fly on American airlines.

c. Please repeat that.

d. Does American 487 have a first class section?

e. I need to fly between Philadelphia and Atlanta.

f. What is the fare from Atlanta to Denver?

g. Is there an American airlines flight from Philadelphia to Dallas?

2. Consider the below grammar to construct the Early parser for the input string "Christ Opened the eyes". (10M) (CO3) (BLT4)

S -> NP VP VP -> V NP PP V -> opened

NP -> N PP -> Prep NP V -> ate

NP -> Det NP N -> Christ Det -> the

NP -> NP NP N -> blind Det -> a

VP -> V N -> eyes Prep -> of

VP -> V NP N -> bread Prep -> in

QUESTION 1 – TREE STRUCTURES FOR ATIS SENTENCES

1.1 Introduction

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand and process human languages. One important task in NLP is syntactic analysis, which identifies the grammatical structure of a sentence.

A sentence contains different grammatical units such as nouns, verbs, adjectives, pronouns, prepositions and noun phrases. These units are arranged according to grammatical rules.

A tree structure is used to represent the grammatical organization of a sentence. It shows how individual words are grouped into phrases and how those phrases combine to form a complete sentence.

In NLP, tree structures are also known as syntax trees or parse trees.

For example:

Sentence: The student reads a book.

The sentence contains:

- A noun phrase: The student.
- A verb phrase: reads a book.
- A complete sentence: The student reads a book.

The tree structure represents these relationships in a hierarchical form.

1.2 Objectives

1. To understand the concept of syntactic analysis.
2. To study the structure of ATIS sentences.
3. To identify different grammatical categories.
4. To construct tree structures for given sentences.
5. To understand the relationship between words and phrases.
6. To represent sentence structure using syntax trees.

1.3 What is ATIS?

ATIS stands for Air Travel Information System.

It is a well-known NLP dataset and task domain used for understanding spoken and written queries related to air travel.

ATIS sentences usually contain questions and commands about:

- Airline information.
- Flight availability.
- Departure and arrival locations.
- Flight timings.

- Ticket fares.
- Passenger requirements.
- Travel classes.
- Airline services.

Examples:

- Does American Airlines have a flight?
- What is the fare from Atlanta to Denver?
- I need to fly between Philadelphia and Atlanta.

These sentences are useful for studying how computers identify grammatical structures and understand travel-related queries.

1.4 What is a Tree Structure?

A tree structure is a hierarchical representation of a sentence.

It begins with a root node representing the complete sentence. The root node is divided into smaller grammatical units until individual words are reached.

A basic syntax tree contains the following components:

Component	Meaning
Root node	Represents the complete sentence
S	Sentence
NP	Noun Phrase
VP	Verb Phrase
PP	Prepositional Phrase
DT	Determiner
NN	Noun
VB	Verb
IN	Preposition
PRP	Personal Pronoun
MD	Modal Auxiliary
WHNP	Wh-Noun Phrase

1.5 Common Grammatical Categories

Short form	Full form	Example
S	Sentence	Complete sentence
NP	Noun Phrase	American Airlines
VP	Verb Phrase	have a flight
PP	Prepositional Phrase	from Atlanta
WHNP	Wh-Noun Phrase	What
PRP	Personal Pronoun	I
VB	Base Verb	fly
VBP	Present Verb	have
VBZ	Third-person singular verb	Does
MD	Modal Verb	would
DT	Determiner	a, the
NN	Singular Noun	flight
NNP	Proper Noun	American
CD	Cardinal Number	487
IN	Preposition	from, between
RB	Adverb	Please
TO	Infinitive Marker	to
CC	Coordinating Conjunction	and

1.6 Basic Structure of a Sentence

A sentence can be divided into two major parts:

\[S \rightarrow NP + VP \]

Where:

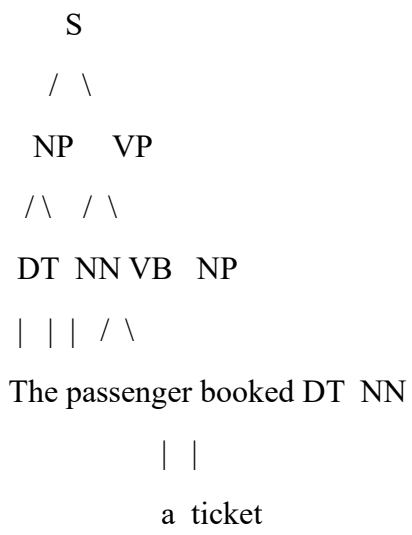
- S = Sentence.
- NP = Noun Phrase.
- VP = Verb Phrase.

Example:

The passenger booked a ticket.

- NP: The passenger.
- VP: booked a ticket.

Basic tree diagram:



1.7 Main Components of Tree Structures

1. Root node

The root node is the topmost node of the syntax tree. It represents the entire sentence.

2. Internal nodes

Internal nodes represent grammatical categories such as NP, VP and PP.

3. Leaf nodes

Leaf nodes represent the individual words of the sentence.

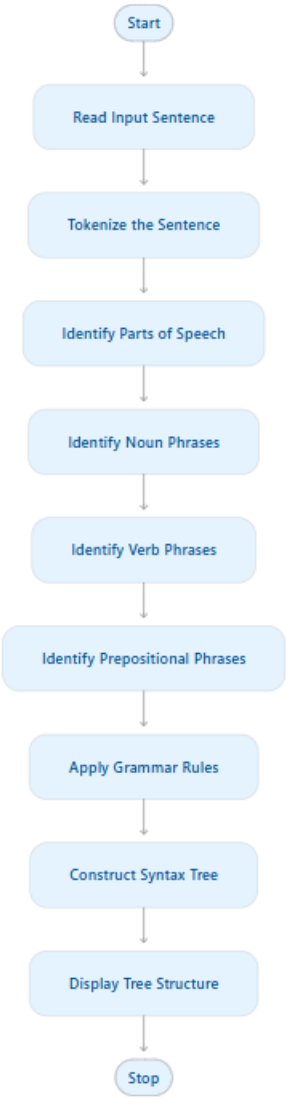
4. Branches

Branches connect the parent nodes with their child nodes.

5. Hierarchical arrangement

The tree represents the grammatical relationship between words and phrases.

1.8 General Flowchart for Constructing a Syntax Tree



1.9 Steps for Constructing a Tree Structure

1. Read the complete sentence.
2. Separate the sentence into individual words.
3. Identify the grammatical category of each word.
4. Find the noun phrase in the sentence.
5. Find the verb phrase in the sentence.
6. Identify any prepositional or other phrases.
7. Connect the phrases according to grammatical rules.
8. Draw the tree from the root node to the individual words.
9. Verify that every word is included in the tree.

1.10 Tree Structure for Sentence (a)

Sentence: Does American airlines have a flight between five a.m. and six a.m.?

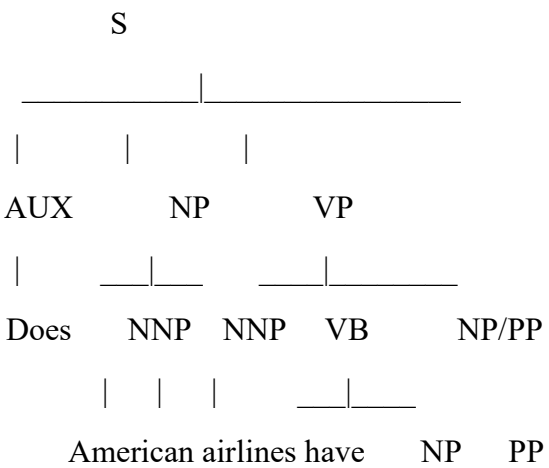
Step 1: Identify the grammatical components

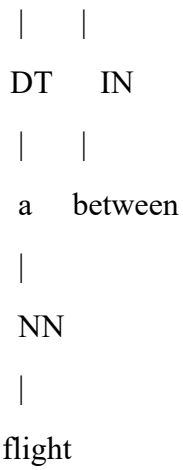
Word	Grammatical category
Does	Auxiliary verb
American	Proper noun
airlines	Proper noun
have	Main verb
a	Determiner
flight	Noun
between	Preposition
five	Number
a.m.	Time expression
and	Conjunction
six	Number
a.m.	Time expression

Step 2: Identify the phrases

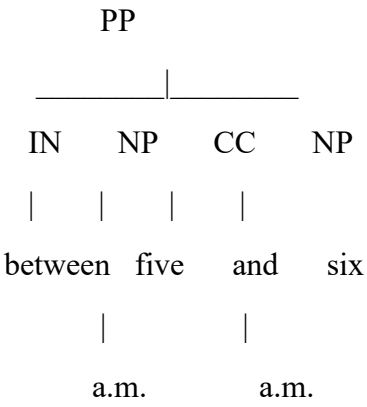
- Auxiliary: Does
- Subject NP: American airlines
- Main VP: have a flight between five a.m. and six a.m.
- Object NP: a flight
- Time PP: between five a.m. and six a.m.

Step 3: Tree diagram





The time expression can be represented as a coordinated prepositional phrase:



Combined grammatical interpretation:

The sentence asks whether American Airlines operates a flight during the specified time interval.

Result: The sentence is divided into an auxiliary verb, subject noun phrase, verb phrase, object noun phrase and time expression.

1.11 Tree Structure for Sentence (b)

Sentence: I would like to fly on American airlines.

Step 1: Identify the grammatical components

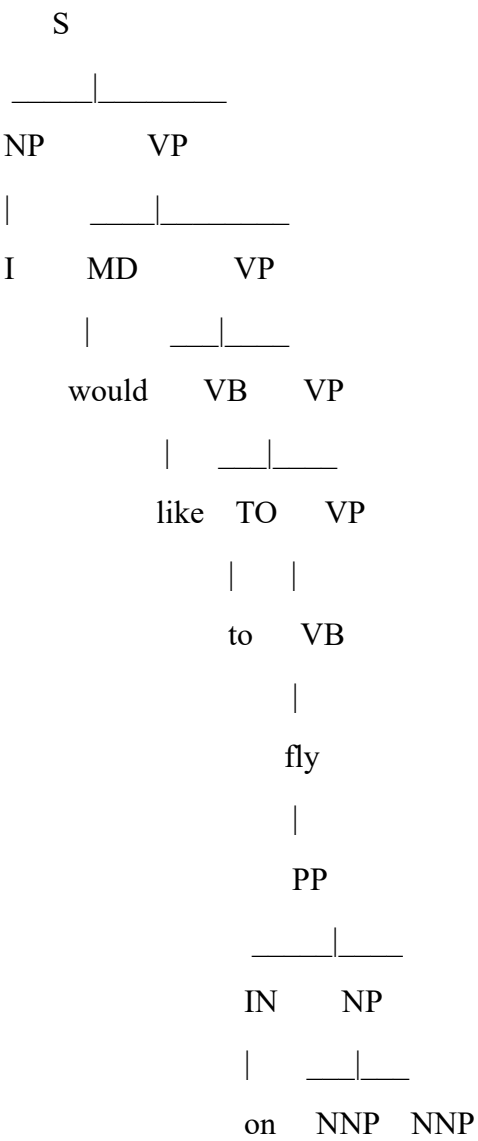
Word	**Grammatical category**
I	Personal pronoun
would	Modal auxiliary
like	Main verb
to	Infinitive marker
fly	Verb

on	Preposition
American	Proper noun
airlines	Proper noun

Step 2: Identify the phrases

- Subject NP: I
- Auxiliary: would
- Main verb: like
- Infinitive phrase: to fly
- PP: on American airlines
- Object of preposition: American airlines

Step 3: Tree diagram





American airlines

Explanation:

- I is the subject of the sentence.
- Would is a modal auxiliary.
- Like is the main verb.
- To fly is an infinitive phrase.
- On introduces a prepositional phrase.
- American airlines is the object of the preposition.

Result: The sentence represents a passenger expressing a preference to travel using American Airlines.

1.12 Tree Structure for Sentence (c)

Sentence: Please repeat that.

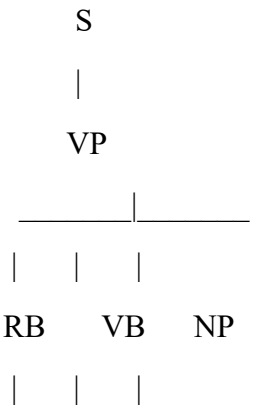
Step 1: Identify the grammatical components

Word	Grammatical category
Please	Imperative politeness marker
repeat	Base verb
that	Demonstrative pronoun

Step 2: Identify the phrases

- Imperative marker: Please
- Main verb: repeat
- Object NP: that

Step 3: Tree diagram



Please repeat PRP

|

that

Explanation:

- Please is used to make the command polite.
- Repeat is the main action.
- That is the object of the verb.
- The sentence is an imperative sentence because it gives a request or instruction.

Result: The sentence is represented by a verb phrase containing a politeness marker, a main verb and an object.

1.13 Tree Structure for Sentence (d)

Sentence: Does American 487 have a first class section?

Step 1: Identify the grammatical components

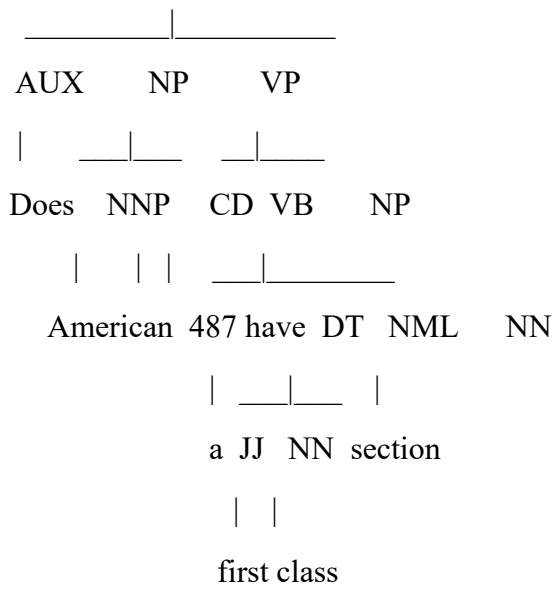
Word	**Grammatical category**
Does	Auxiliary verb
American	Proper noun
487	Cardinal number
have	Main verb
a	Determiner
first	Ordinal adjective
class	Noun modifier
section	Singular noun

Step 2: Identify the phrases

- Auxiliary: Does
- Subject NP: American 487
- Main verb: have
- Object NP: a first class section

Step 3: Tree diagram

S



Explanation:

- Does is an auxiliary verb used to form a question.
- American is a proper noun.
- 487 is the flight number.
- Have is the main verb.
- A is a determiner.
- First and class together describe the type of section.
- Section is the main noun of the object phrase.

Result: The tree shows a question about the availability of a first-class section on flight American 487.

1.14 Tree Structure for Sentence (e)

Sentence: I need to fly between Philadelphia and Atlanta.

Step 1: Identify the grammatical components

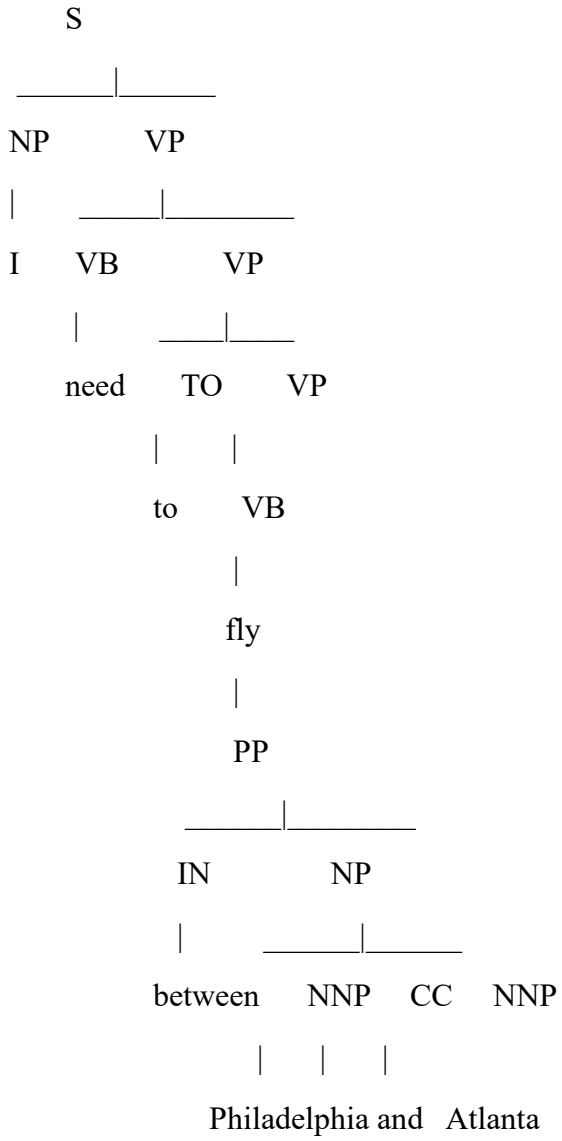
Word	**Grammatical category**
I	Personal pronoun
need	Main verb
to	Infinitive marker
fly	Base verb
between	Preposition

Philadelphia	Proper noun
and	Conjunction
Atlanta	Proper noun

Step 2: Identify the phrases

- Subject NP: I
- Main verb: need
- Infinitive phrase: to fly
- PP: between Philadelphia and Atlanta
- Coordinated locations: Philadelphia and Atlanta

Step 3: Tree diagram



Explanation:

- I is the subject.
- Need expresses the passenger's requirement.
- To fly forms an infinitive phrase.
- Between introduces the two travel locations.
- Philadelphia and Atlanta are connected using the conjunction and.

Result: The sentence represents a travel requirement between two cities.

1.15 Tree Structure for Sentence (f)

Sentence: What is the fare from Atlanta to Denver?

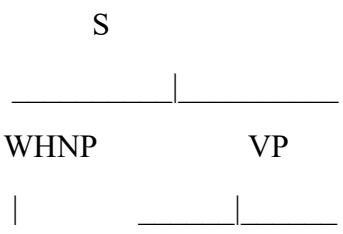
Step 1: Identify the grammatical components

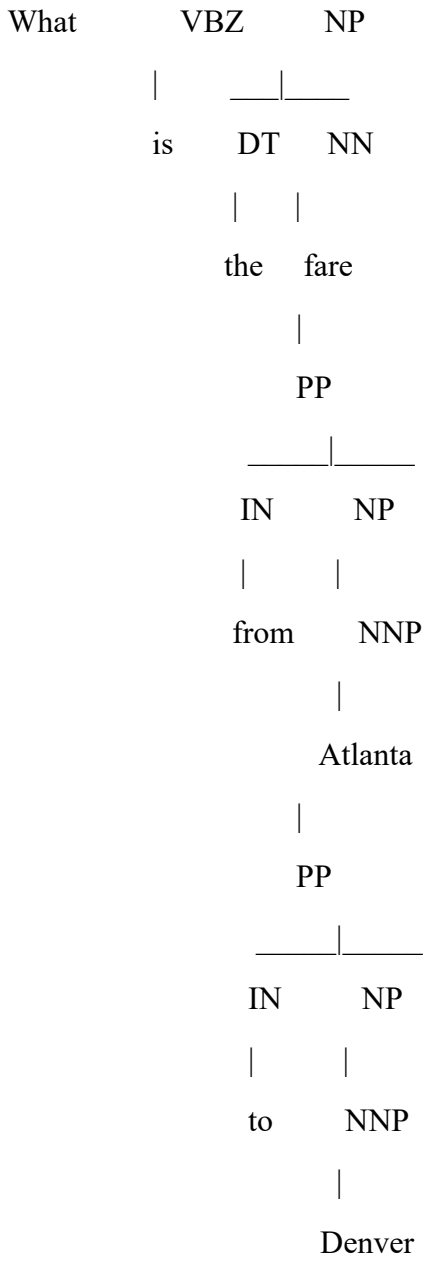
Word	**Grammatical category**
What	Wh-pronoun
is	Auxiliary/linking verb
the	Determiner
fare	Noun
from	Preposition
Atlanta	Proper noun
to	Preposition
Denver	Proper noun

Step 2: Identify the phrases

- Wh-NP: What
- Verb: is
- Subject NP: the fare
- PP: from Atlanta
- PP: to Denver

Step 3: Tree diagram





Explanation:

- What introduces a question about the fare.
- Is is the linking verb.
- The fare is the main noun phrase.
- From Atlanta identifies the starting location.
- To Denver identifies the destination.

Result: The tree represents a question asking for the ticket price between Atlanta and Denver.

1.16 Tree Structure for Sentence (g)

Sentence: Is there an American airlines flight from Philadelphia to Dallas?

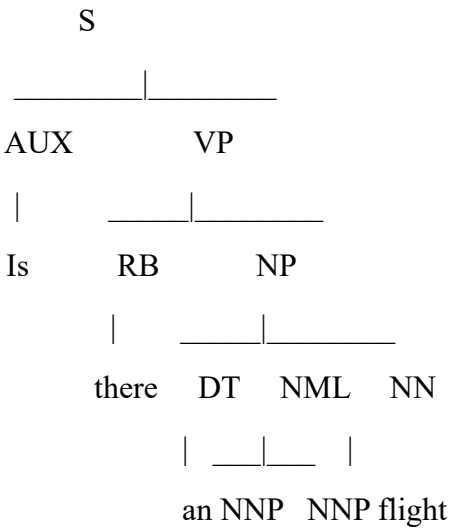
Step 1: Identify the grammatical components

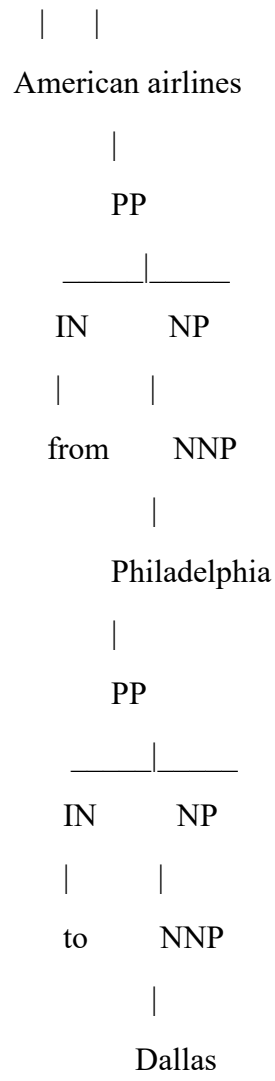
Word	**Grammatical category**
Is	Auxiliary/linking verb
there	Existential adverb
an	Determiner
American	Proper noun
airlines	Proper noun
flight	Noun
from	Preposition
Philadelphia	Proper noun
to	Preposition
Dallas	Proper noun

Step 2: Identify the phrases

- Auxiliary: Is
- Existential expression: there
- Subject/NP: an American airlines flight
- PP: from Philadelphia
- PP: to Dallas

Step 3: Tree diagram





Explanation:

- Is is used to form a yes/no question.
- There introduces an existential construction.
- An is a determiner.
- American airlines identifies the airline.
- Flight is the main noun.
- From Philadelphia and to Dallas specify the departure and destination locations.

Result: The sentence asks whether an American Airlines flight is available from Philadelphia to Dallas.

1.17 Consolidated Table of All Seven Sentences

Sentence	**Main structure**	**Main phrase components**

(a) Does American airlines have a flight between five a.m. and six a.m.?	Question	AUX + NP + VP
(b) I would like to fly on American airlines.	Statement	NP + VP + PP
(c) Please repeat that.	Imperative	VP + NP
(d) Does American 487 have a first class section?	Question	AUX + NP + VP
(e) I need to fly between Philadelphia and Atlanta.	Statement	NP + VP + PP
(f) What is the fare from Atlanta to Denver?	Wh-question	WHNP + VP
(g) Is there an American airlines flight from Philadelphia to Dallas?	Yes/no question	AUX + VP + NP + PP

1.18 Importance of Tree Structures in NLP

1. Tree structures show how words are arranged in a sentence.
2. They help identify noun phrases and verb phrases.
3. They represent grammatical relationships between words.
4. They help computers understand natural-language questions.
5. They are useful for sentence parsing and language understanding.
6. They support information extraction from user queries.
7. They help identify the meaning and structure of complex sentences.

1.19 Applications of Syntax Trees

1. Used in machine translation.
2. Used in grammar checking systems.
3. Used in question-answering applications.
4. Used in voice-based travel assistants.
5. Used in text summarization.
6. Used in information extraction.
7. Used in chatbot development.

1.20 Advantages and Limitations

Advantages

1. Provides a clear representation of sentence structure.
2. Helps identify grammatical categories.
3. Makes sentence analysis easier.

4. Supports advanced NLP applications.
5. Helps identify relationships between different phrases.

Limitations

1. Long sentences may create complex trees.
2. Ambiguous sentences may have multiple possible trees.
3. Informal expressions may be difficult to analyze.
4. Proper nouns and compound phrases need careful treatment.
5. Different grammar rules may produce different tree structures.

1.21 Result

Tree structures were constructed for all seven ATIS sentences. Each sentence was divided into grammatical components such as noun phrases, verb phrases, prepositional phrases, auxiliary verbs and other word categories. The syntax trees demonstrate how NLP identifies the grammatical organization of air-travel-related sentences.

QUESTION 2 – EARLEY PARSER FOR THE INPUT SENTENCE

1. Introduction

The Earley Parser is a parsing algorithm used in Natural Language Processing to check whether a sentence follows a given grammar. It uses three operations: Prediction, Scanning and Completion.

2. Given Grammar

1. S → NP VP
2. VP → V NP
3. VP → V
4. PP → Prep NP
5. V → opened
6. V → ate
7. NP → N
8. NP → Det NP
9. NP → NP NP
10. N → Christ
11. N → blind
12. N → eyes

13. N → bread
14. Det → the
15. Det → a
16. Prep → of
17. Prep → in

3. Input Sentence

Christ Opened the Eyes

Tokens: Christ | Opened | the | eyes

4. Earley Parser Operations

- Prediction: Adds possible grammar rules for the expected symbol.
- Scanning: Matches the next input word with a grammar terminal.
- Completion: Combines completed rules with earlier waiting rules.

5. Algorithm

1. Initialize the chart with the starting rule S' → S.
2. Apply prediction to find possible grammar rules.
3. Apply scanning to match input words.
4. Apply completion when a grammar rule is finished.
5. Repeat until all input words are processed.
6. Check whether the final starting rule is completed.

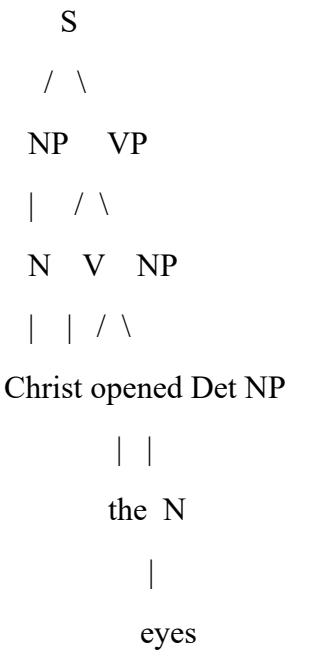
6. Parsing Steps

The input is converted to lowercase for matching with the grammar.

Chart	Input position	Important operation
C0	Before Christ	Predict NP and N
C1	After Christ	Scan Christ and complete NP
C2	After Opened	Scan opened and complete V
C3	After the	Scan the and predict NP
C4	After eyes	Scan eyes and complete NP and VP

The parser successfully recognizes the complete sentence.

7. Parse Tree



8. Python Program

```
grammar = {
    "S": [["NP", "VP"]],
    "VP": [["V", "NP"], ["V"]],
    "NP": [["N"], ["Det", "NP"], ["NP", "NP"]],
    "N": [["christ"], ["blind"], ["eyes"], ["bread"]],
    "V": [["opened"], ["ate"]],
    "Det": [["the"], ["a"]],
    "PP": [["Prep", "NP"]],
    "Prep": [["of"], ["in"]]
}
sentence = "Christ Opened the Eyes".lower().split()
def earley_parse(words):
    chart = [set() for _ in range(len(words) + 1)]
    chart[0].add(("START", ("S",), 0, 0))
    for i in range(len(words) + 1):
        changed = True
        while changed:
            changed = False
```

```
            for state in list(chart[i]):
                lhs, rhs, dot, start = state
                if dot < len(rhs):
                    symbol = rhs[dot]
                    if symbol in grammar:
                        for rule in grammar[symbol]:
                            new_state = (symbol, tuple(rule), 0, i)
                            if new_state not in chart[i]:
                                chart[i].add(new_state)
                                changed = True
                    elif i < len(words) and symbol == words[i]:
                        chart[i + 1].add(
                            (lhs, rhs, dot + 1, start)
                        )
                else:
                    for old in list(chart[start]):
                        a, b, d, s = old
                        if d < len(b) and b[d] == lhs:
                            new_state = (a, b, d + 1, s)
                            if new_state not in chart[i]:
                                chart[i].add(new_state)
                                changed = True
    return ("START", ("S",), 1, 0) in chart[len(words)]
if earley_parse(sentence):
    print("Sentence is accepted")
else:
    print("Sentence is rejected")
```

OUTPUT:

```
Sentence is accepted
```

10. Applications

1. Used in natural language processing.
2. Helps analyze sentence structures.
3. Used in speech recognition.
4. Supports grammar checking.
5. Helps build language understanding systems.

11. Advantages

1. Handles different grammar structures.
2. Supports ambiguous sentences.
3. Uses prediction, scanning and completion.
4. Works with context-free grammars.

12. Conclusion

The Earley Parser successfully analyzes the sentence “Christ Opened the Eyes” using the given grammar. It identifies the sentence structure through prediction, scanning and completion.